

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			The number of species and the number of individuals of each species in a (given) {environment/ habitat/ ecosystem/ area}/ species richness and (species) evenness in an environment (1)	1			1		
	(b)			(>11) no increase in numbers of species present (so waste of time)/ maximum cumulative species total (1) (<11){not all/ fewer} species present/ sample not representative (1)		2		2		2
	(c)	(i)		<u>Erica</u>		1		1		
		(ii)		Only {presence/ absence} of a species recorded/ the number of {individuals/ a species} is not recorded some quadrats may contain >1 of a species (1)			1	1		2
		(iii)		D = 0.77 = 3 marks If incorrect award 2 marks for: 0.77183600 /0.772 (correct calculation but not to 2 dp) 0.8 (correct calculation but not to 2 dp) 0.78 (incorrect rounding but given to 2 dp) 0.23 (correct calculation but not subtracted from 1) If incorrect award 1 mark for either of $\sum n(n-1) = 256$ $N(N-1) = 1122$ 0.771 (correct calculation but not to 2 dp and not correctly rounded)		3		3	3	3
	(d)	(i)		Grazing increases biodiversity Ecf if answer to (iii) is > 0.87 then biodiversity decreases			1	1		1

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		(ii)		Any two (x1) from - Grazing reduces the growth of {medium plants/ tall plants/ silver birch/ gorse/ brambles} (1) - More {light/ nutrients/ water/ named nutrients} available/ decreased competition for {light/ nutrients/ water/ named nutrients} (1) - Grazers provide fertilizer for plants/ owtte (1) {Seed/ pollen} dispersal/ owtte (1)			2	2		
				Question 1 total	1	6	4	11	3	8